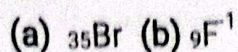


Task1:

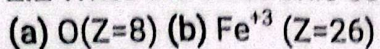
1. Fill in the blank with (✓) in the table of Atomic Models Development below:

The Atom Model	Indivisible	Electron	Nucleus	Orbit	Neutron	Electron Cloud
Dalton	✓					
Chadwick						
Thomson		✓				
Rutherford		✓	✓			✓
Bohr		✓	✓	✓		
Wave (modern model)		✓	✓		✓	✓

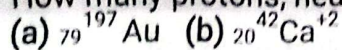
2.1 Write the orbital notation for:



2.2 Write the electronic configuration by the simplified notation for:



3. How many protons, neutrons, and electrons are in:

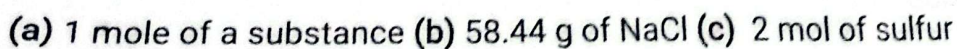


4. Complete the following table:

Element	Atomic Number	Atomic Mass (amu)	Protons	Neutrons	Electrons	Sample mass (g)	No. of moles
F	9	19.00	9	10	9	13.3	0.7
Na	11	22.99	11	12	11	3	0.130
S	16	32.07	16	16	16	8	0.250

I change this

5. How many atoms are there in...



By written Students

1) Gubran Mohammed Ghaleb Mohammed $\rightarrow 2505072095$

2) Mohammed Amin Hassan Ali 2505072201

Q5: How many atoms are there in...

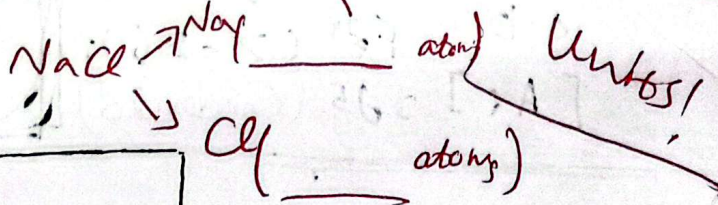
(a): 1 mole $\times 6.022 \times 10^{23} = \boxed{6.022 \times 10^{23}}$

(b): 58.44g NaCl

$\cancel{\text{mol}} \times \frac{\text{atom}}{\cancel{\text{mol}}}$

(7)

$\frac{58.44}{58.44} = 1 \text{ mol}$



$2 \times 6.022 \times 10^{23} = 1.204 \times 10^{24}$

$1 \times 6.022 \times 10^{23} = 6.022 \times 10^{23}$

g/mol

c: 2 moles $\Rightarrow 2 \times 6.022 \times 10^{23} = 1.204 \times 10^{24}$

d: 12.01g $\frac{12.01}{12.01} = 1 \text{ mol}$ (g/mol)

$1 \times 6.022 \times 10^{23} = \boxed{6.022 \times 10^{23}}$

Units!

e: 300 mg R

$\frac{0.3}{39.10} \approx 0.0076 \text{ mol}$

$0.00767 \times 6.022 \times 10^{23} \approx \boxed{4.62 \times 10^{21}}$

Units!

278 (a) Total Mass of 4 moles of (Mg)

$4 \text{ mol} \times 24.305 \text{ g/mol} = 97.22 \text{ g}$ (4)

(b) Total number of Atoms

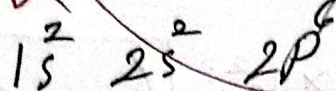
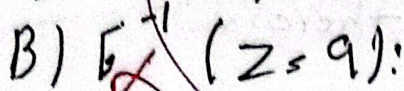
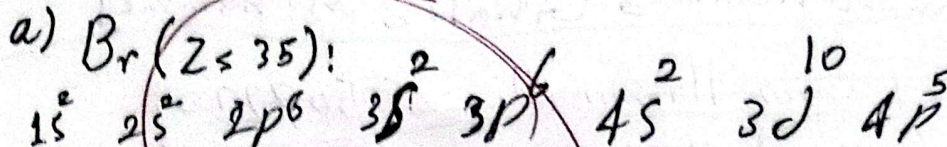
$4 \text{ mol} \times 6.022 \times 10^{23} \text{ atoms/mol} = 2.409 \times 10^{24} \text{ atoms}$

Mohammed Amin Hassan Ali
2505072201

Q 2.1

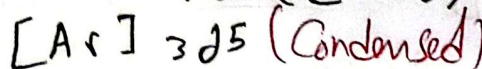
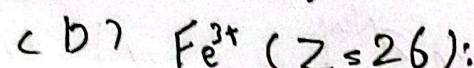
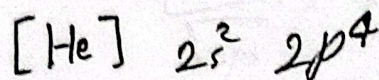
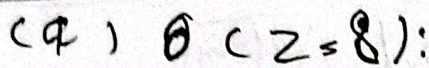
①

Orbital notation



Q 2.2

⑤

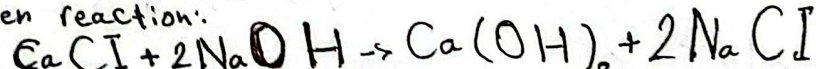


Khath Osama Al: AWath
2505072158

Q 8)

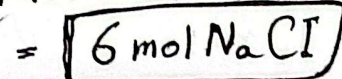
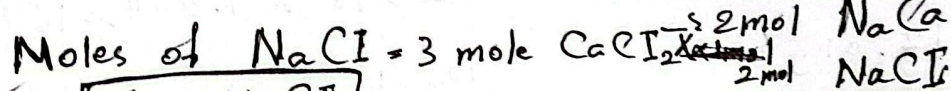
Reaction calculations

Given reaction:



a) Moles of NaCl Produced from 3 moles of $CaCl_2$

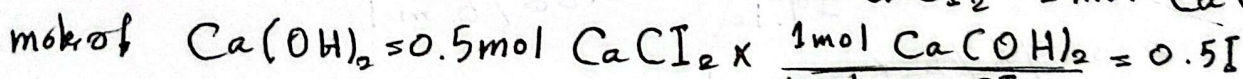
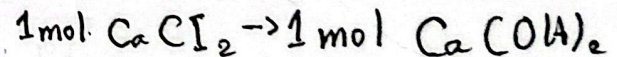
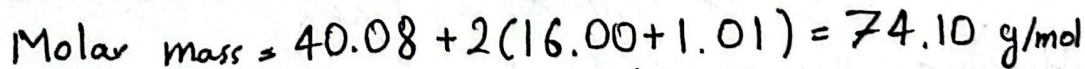
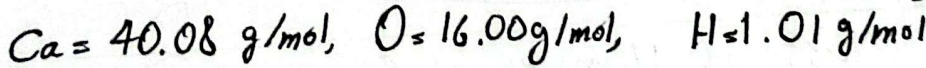
from stoichiometry: 1 mole $CaCl_2$



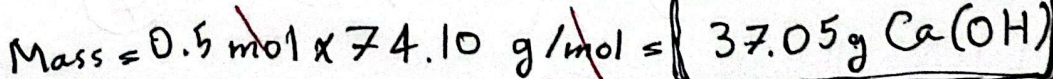
b) Mass of $Ca(OH)_2$ Produced from 0.5 moles of $CaCl_2$

• Sol

1. Stoichiometry:

2. Molar mass of $Ca(OH)_2$:

3. Mass calculation:



Mohammed Mazin
2505072157

~~Fill in the blank with its in the table of average~~
~~models Development are below~~

~~The atom model~~

~~The atom model~~ Indivisible 1. Electron - nucleon

3:13

Name: Abdullah Fathi Mohammed Saleh

Reg No: 2505072084

a) K_2SO_4 : Potassium Sulphate

b) H_2SO_3 : Sulfurous acid

c) $FeCl_3$: Iron(III) chloride

d) $Cu(NO_3)_2$: Copper(II) nitrate

5

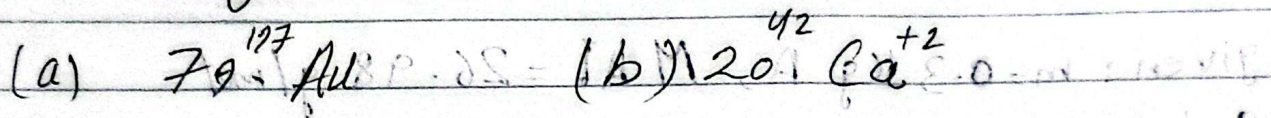
Q. 14 ?! (Zero)

$\frac{85}{100}$

$\frac{17}{20}$

* Mona Khaleel Abdelrahman Fadel (2505072023)

* ③ How many Protons, neutrons, and electrons are in:



(a) Gold - $^{197}_{79}\text{Au}$: (F)

- Protons = Atomic number (Z) = 79
- Neutrons = Mass number (A) - Z = 197 - 79 = 118
- Electrons = Protons (neutral atom) = 79

(b) Calcium - $^{42}_{20}\text{Ca}^{+2}$:

- Protons = Atomic number (Z) = 20
- Neutrons = 42 - 20 = 22
- Electrons = Protons - charge = 20 - 2 = 18

* Shada Sami Hamed Mohammed (2505072003)

④. Complete the following table: (G)

Element	Atomic Number	Atomic Mass	Protons	Neutrons	Electrons	Sample Mass	No. of Moles
F	9	19.00	9	10	9	13.3	0.7
Na	11	22.99	11	12	11	3	0.130
S	16	32.07	16	16	16	8	0.250

* Haneen Talaat Abdul Malek Yaseen (2505072004)

* ⑥ Moles of Aluminum (Al) in 0.3 Kg

given: $m = 0.3 \text{ Kg Al}$, $M_{\text{Al}} = 26.98 \text{ g/mol}$

Solution:-

1. convert mass to grams:-

$$0.3 \text{ Kg} \times 1000 \text{ g/Kg} = 300 \text{ g}$$

2. calculate mole:-

$$n = \frac{m}{M} = \frac{300 \text{ g}}{26.98 \text{ g/mol}}$$

$$n \approx 11.12 \text{ mol}$$

$$11.12 \text{ mol}$$

* Ayat shihab Ahmed Mahdi (2505072036)

9) Write the Following numbers using the proper Scientific notation:

① 9,876,000

② 0.0004567

③ 546.6×10^3

(a) $9,876,000 \Rightarrow 9.87 \times 10^6$

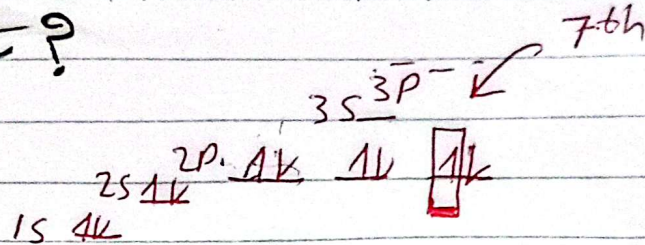
(b) $0.0004567 \Rightarrow 4.567 \times 10^{-4}$

(c) $546.6 \times 10^3 \Rightarrow 5.466 \times 10^5$

Farah Jalal Abdul-Jabbar Abdul-Grafoor Khan (2505072040).

IV. Give the four quantum numbers for the seventieth electron in $_{14}\text{Si}$ element?

- $n = 2$
- $l = 1$
- $m_l = 0$
- $m_s = +\frac{1}{2}$



(4)

* Rawan Saeed Nasser Mohammed Barashed .
(2505072181)

[12] Determine whether each of the following bonds will be ionic, polar covalent, or nonpolar covalent :-

- (a) Cl (3.0) and Na (1.0) (b) H (2.1) and O (3.5)
(c) N (3.0) and P (2.1) (d) C (2.5) and H (2.1)

a. Cl - Na: Ionic ($\Delta EN = 2.0$). (9)

b. H - O: Polar covalent ($\Delta EN = 1.4$).

c. N - P: Polar covalent ($\Delta EN = 0.9$).

d. C - H: Non polar covalent ($\Delta EN = 0.4$).

2505072120

Amir

Abdullah Qassem Abdulaziz Abdullah

Fill in the blank with (✓) in the table of Atomic models

Development below

The Atom model	Indivisible	Electron	nucleus	orbit	neutron	Electron cloud
Dalton	✓					
Chadwick						
Thomson		✓				
Rutherford		✗	✓			
Bohr		✗	✗	✓		
wave (modern model)		✗	✗		✗	✓

Q 10:

Sol

Name: Ammar Mohammed Al-Khatib

Reg: 2505072075.

(8)

$$(a) (15.2 + 3.45) \times 2.5$$

$$15.2 + 3.45 = 18.65 \rightarrow 18.7$$

$$18.7 \times 2.5 = 46.75 \rightarrow \boxed{47} \text{ (2 sig fig.)}$$

$$(b) (48.7 - 12.5) \div 3.1$$

$$48.7 - 12.5 = 36.2$$

$$36.2 \div 3.1 = 11.677 \rightarrow \boxed{12} \text{ (2 sig. fig.)}$$